

Detecting Structural Breaks in Crypto Markets

A beginner-friendly empirical report inspired by structural-break methods in
Advances in Financial Machine Learning

Prepared from Binance daily OHLC market data

Generated on 2026-07-08

What this report does. We take historical crypto price data for BTC, ETH, ETC, SOL, and HYPE and apply structural-break diagnostics: a BDE-inspired recursive forecast-error CUSUM, Chu-Stinchcombe-White CUSUM, Chow-type Dickey-Fuller, SADF, QADF, a conditional-tail ADF summary, and sub/super-martingale trend tests (SM-Exp, SM-Power, SM-Poly1, SM-Poly2), plus a walk-forward Gaussian hidden Markov model (HMM) regime filter. The goal is educational: show how the tests behave, how to interpret the plots, and why one should treat flagged regimes as research features rather than automatic trading signals.

Contents

1	Intuition	3
2	Data and Parameters	3
3	Diagnostic Families	4
3.1	CUSUM Tests	4
3.2	ADF-Style Tests	5
3.3	SMT-Style Trend Tests	5
3.4	Hidden Markov Regime Model	5
4	Formal Inference Versus Research Features	6
5	How to Read the Plots	6
6	Results by Asset	6
6.1	BTC	6
6.2	ETH	7
6.3	ETC	9
6.4	SOL	10
6.5	HYPE	12

7	Consensus Alert Periods	14
8	Finite-Sample Calibration and Sensitivity	15
9	Long/Flat Backtests	16
9.1	Backtest Rule	16
9.2	PnL Curves	17
9.3	Full Strategy Table	19
9.4	Walk-Forward and Multiple-Testing Checks	20
10	Interpretation	21
11	Minimal Reproducibility Sketch	21
12	Reproduce	21
13	References	22

1 Intuition

A **structural break** is a change in the data-generating mechanism. In markets this often means that a process that looked stable, noisy, or mean-reverting starts behaving like a momentum process, an explosive bubble, or a fast collapse. These breaks can be useful because many market participants adapt slowly, but the diagnostics are not crash predictors. A strong reading says: *something about the recent price path is unusual under the reference model*. That unusual behavior may be a bubble, a burst, a volatility shock, a liquidity episode, a listing effect, a leverage unwind, or a false alarm.

2 Data and Parameters

The raw source data are Binance daily OHLCV klines. The report uses the largest available Binance spot USDT history where spot is available. HYPEUSDT is not available on Binance spot in the API check used here, so HYPE falls back to Binance USD-M futures daily klines. All diagnostics are computed on daily closes through 2026-07-07.

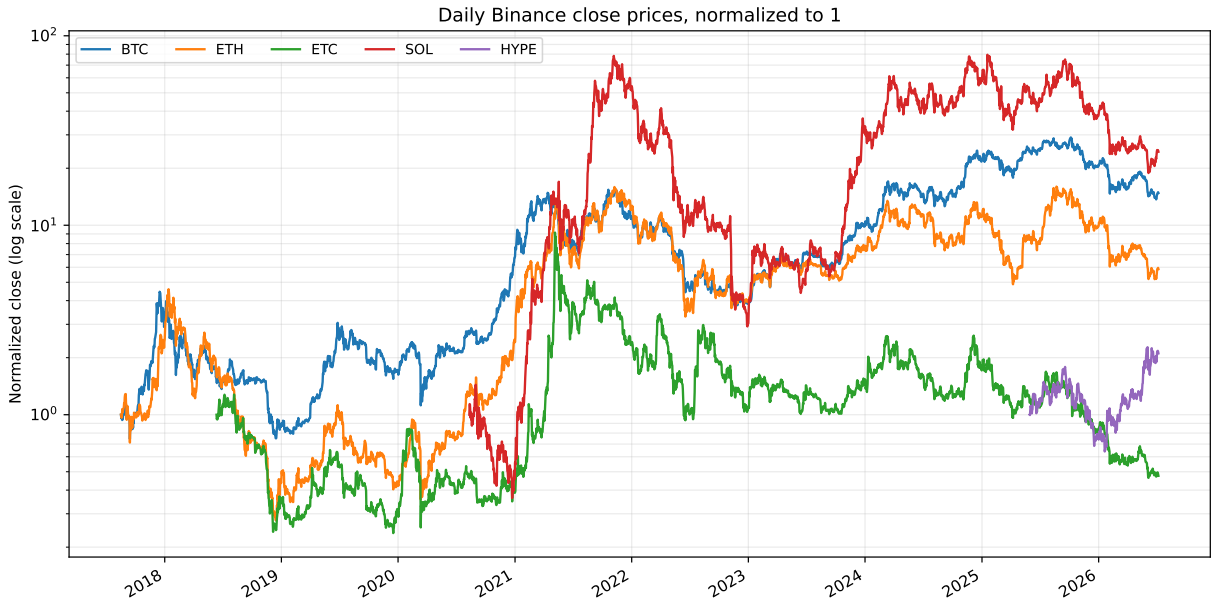


Figure 1: Downloaded daily close prices, normalized to 1 at each asset’s first available row. A log scale is used because crypto prices can move by orders of magnitude.

Asset	Source	Raw start	Raw end	Raw rows	τ	SDFC candidate	SDFC t	Max SADF	Max CSW	Max SMT	Runs
BTC	Binance spot	2017-08-17	2026-07-07	3247	488	2019-02-08	1.38	2021-01-08	2020-03-12	2025-10-28	16
ETH	Binance spot	2017-08-17	2026-07-07	3247	488	2020-03-13	0.90	2021-05-11	2020-03-12	2021-12-09	18
ETC	Binance spot	2018-06-12	2026-07-07	2948	443	none (non-positive t)	-0.26	2021-05-06	2020-03-12	2021-11-17	6
SOL	Binance spot	2020-08-11	2026-07-07	2157	324	2022-12-30	0.59	2023-12-25	2022-11-09	2022-01-04	10
HYPE	Binance USD-M futures	2025-05-30	2026-07-07	404	61	2026-01-21	1.71	2026-06-01	2026-01-28	2025-12-28	7

Table 1: Daily Binance data and headline diagnostic results. τ is the minimum daily window length used in recursive tests. Runs count retrospective consensus alert runs, where at least two diagnostic families were simultaneously in their research-alert region.

Parameter	Value
Transaction cost	0.10% per exposure change
Online signal quantile	0.95 prior expanding quantile
Minimum online history	60 finite observations
Momentum gate	trailing 20-day log return must be positive
Holding window	20 calendar daily observations after an alert
Minimum diagnostic window	$\max(60, \lceil 0.15n \rceil)$
HMM regime model	3 states, refit every 126 obs, min 365 training obs
ADF Monte Carlo calibration	2000 random-walk simulations, seed 123
Walk-forward validation	expanding 730-day train, 180-day test blocks

Table 2: Backtest and diagnostic parameters used by the generated report.

3 Diagnostic Families

The table below maps each statistic in this report to the source material it is based on, so readers can connect the empirical output back to the original definitions.

Report statistic	Source	Fidelity
BDE-inspired CUSUM	AFML §17.3.1 (Brown-Durbin-Evans 1975)	inspired, online variant
Two-sided CSW excess	AFML §17.3.2 (Chu-Stinchcombe-White 1996)	two-sided heuristic
SDFC	AFML §17.4.1 (Chow-type Dickey-Fuller)	faithful, zero-lag
SADF	AFML §17.4.2 (Phillips-Wu-Yu 2011)	faithful, zero-lag
QADF / CADF	AFML §17.4.2.4–17.4.2.5	faithful, zero-lag
SMT score	AFML §17.4.3 (SM-Exp/Power/Poly1/Poly2)	all four forms, one φ
HMM regime filter	Hamilton (1989); Cartea et al. Ch. 12	walk-forward, filtered

Table 3: Where each diagnostic comes from and how closely it follows the source definition.

3.1 CUSUM Tests

CUSUM means cumulative sum. The BDE-inspired series in this report is a recursive one-step forecast-error CUSUM from a simple log-price trend model. At each close t the trend model is refitted on all data before t , and the standardized forecast error

$$\hat{\omega}_t = \frac{y_t - \hat{\alpha}_{t-1} - \hat{\beta}_{t-1}t}{\hat{\sigma}_{t-1}\sqrt{1 + h_t}},$$

where h_t is the regression leverage of the new point, is accumulated into a scaled running sum. It is **not** a formal Brown-Durbin-Evans boundary test: residuals are normalized online with shifted expanding statistics so a value at close t uses only information available through t .

The Chu-Stinchcombe-White statistic asks whether log price has moved too far from an earlier reference level relative to realized volatility:

$$S_{n,t} = \frac{y_t - y_n}{\hat{\sigma}_t\sqrt{t - n}}, \quad \hat{\sigma}_t^2 = \frac{1}{t - 1} \sum_{i=2}^t (\Delta y_i)^2.$$

The report applies the level-departure statistic symmetrically to bubbles and crashes. Because of that absolute-value transformation, the displayed CSW excess should be read as a two-sided heuristic rather than the original one-sided 5% test.

3.2 ADF-Style Tests

SDFC scans for a single full-sample unknown break into an explosive alternative. It is not traded here because choosing the break date from the full sample would use future data. SADF, QADF, and CADF scan repeated backward windows ending at each date. The implementation uses the zero-lag ADF form for readability and speed:

$$\Delta y_t = \alpha + \beta y_{t-1} + \varepsilon_t.$$

For formal interpretation, the report now adds cached finite-sample random-walk simulations for SADF/QADF/CADF critical values. The full-sample top 5% plot flags remain research visualizations, not p-values.

3.3 SMT-Style Trend Tests

The sub/super-martingale score evaluates the four trend specifications from the source text: an exponential trend on log price (SM-Exp), a power trend of log price on log time (SM-Power), and quadratic trends on normalized raw price (SM-Poly1) and log price (SM-Poly2). For each specification the trend coefficient $\hat{\beta}_{t_0,t}$ is estimated on backward windows and the penalized absolute t-statistic is aggregated as

$$\text{SMT}_t = \sup_{t_0 \in [1, t-\tau]} \left\{ \frac{|\hat{\beta}_{t_0,t}|}{\hat{\sigma}_{\beta_{t_0,t}} (t - t_0)^\varphi} \right\}, \quad \varphi = 0.5,$$

where the penalty $(t - t_0)^\varphi$ prevents long windows from dominating through mechanically large t-values. High values flag unusually strong trend-like growth or collapse. They are useful features, but the reported score is a compact heuristic rather than a complete production SMT battery across many φ values.

3.4 Hidden Markov Regime Model

In addition to the structural-break statistics, the report fits a Gaussian hidden Markov model (HMM) on two causal daily features: the one-day log return and a trailing multi-day log momentum. The HMM is a different kind of regime tool: instead of testing for a break against a reference model, it assumes the market switches between a small number of latent states s_t following a Markov chain with transition matrix A , and that the observed feature vector is drawn from a state-specific Gaussian,

$$x_t \mid s_t = k \sim \mathcal{N}(\mu_k, \Sigma_k), \quad \Pr[s_t = k \mid x_{1:t}] \propto \left(\sum_j \Pr[s_{t-1} = j \mid x_{1:t-1}] A_{jk} \right) \mathcal{N}(x_t; \mu_k, \Sigma_k).$$

The right-hand recursion is the *filtered* state probability: it conditions only on observations up to t . This latent-state view goes back to Markov-switching models in econometrics and connects to the Markov-chain regime models used for order-flow states in the algorithmic-trading literature.

The implementation is designed so that no forward-looking information can reach the trading signal:

- Parameters are re-estimated walk-forward on an expanding window at fixed refit dates; the model used at close t was fitted only on data before its refit date, which is at or before t .
- The regime at close t is the argmax of *filtered* state probabilities from a causal forward pass over observations up to t . Smoothed (forward-backward) probabilities are never used out of sample, because smoothing conditions on future observations.
- The mapping from state index to bullish/non-bullish is computed from the training window only, by the state-weighted mean of same-window daily returns. If no state has a positive mean return, the signal stays flat.

- Like every other strategy in this report, the position is shifted by one day before it trades.

Assets with less history than the minimum training window (currently HYPE) remain flat and report an inactive HMM. The EM fit uses deterministic quantile-based initialization, so results are exactly reproducible without random seeds.

4 Formal Inference Versus Research Features

Statistic	Threshold type	Formal p-value?	Online?
BDE-inspired CUSUM	shifted expanding residual z-score	no	yes for strategy, no for retrospective plot flags
Two-sided CSW excess	book-inspired curve applied symmetrically	heuristic	yes
SADF/QADF/CADF visual flags	full-sample top 5%	no	no
SADF/QADF/CADF critical exceedances	simulated random-walk finite-sample critical values	approximate	no, retrospective inference only
SDFC	full-sample unknown-break scan	no in this report	no
HMM regime filter	walk-forward Gaussian HMM, filtered (causal) state probabilities	no	yes
Long/flat strategies	prior expanding thresholds plus one-day execution shift	no	yes

Table 4: Interpretation labels for the report’s diagnostics and strategy conversions.

5 How to Read the Plots

Each asset figure has six panels: price, BDE-inspired recursive CUSUM, two-sided CSW excess, ADF-family statistics, SMT-style trend score, and the filtered HMM regime state. Orange shading marks **retrospective consensus alert periods**: at least two diagnostic families were in their full-sample top-alert region or exceeded the fixed CSW boundary. This shading is useful for visual structure clustering, but it is not an online trading signal. In the HMM panel, green shading marks the days on which the filtered bull state was active; unlike the orange shading, this *is* the online strategy state.

The online strategy signals are separate. BDE, SADF, QADF, CADF, and SMT strategy events use prior expanding thresholds only. CSW strategy events use the current positive excess because the statistic already embeds its reference boundary. The HMM strategy holds a long position exactly while the filtered bull state is active. The filtered HMM state per day is also stored in each asset’s diagnostics CSV.

6 Results by Asset

6.1 BTC

16 retrospective consensus alert run(s) were flagged on the daily clock. The first listed run begins on 2021-01-02. The SDFC candidate is 2019-02-08 with t-statistic 1.38. SADF peaks on 2021-01-08, the two-sided CSW heuristic peaks on 2020-03-12, and the SMT score peaks on 2025-10-28. The walk-forward HMM covers 2862 daily observations from 2018-09-06 and spends 749 of them in its bull state.

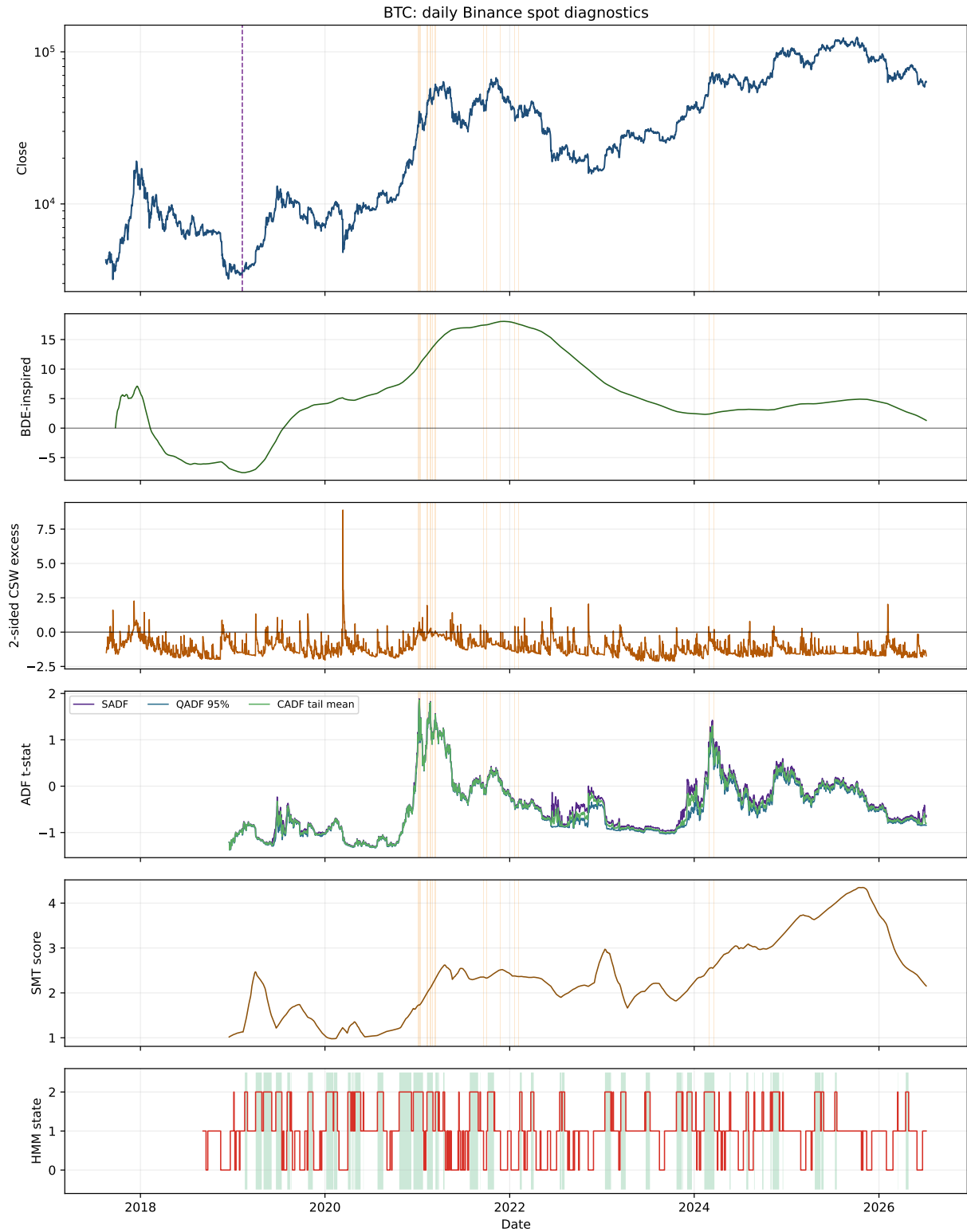


Figure 2: BTC structural-break diagnostics. Orange shading is retrospective consensus, not an online trading signal. Green shading in the bottom panel marks the filtered HMM bull state, which is the online HMM strategy state.

6.2 ETH

18 retrospective consensus alert run(s) were flagged on the daily clock. The first listed run begins on 2021-01-04. The SDFC candidate is 2020-03-13 with t-statistic 0.90. SADF peaks on 2021-05-11,

the two-sided CSW heuristic peaks on 2020-03-12, and the SMT score peaks on 2021-12-09. The walk-forward HMM covers 2862 daily observations from 2018-09-06 and spends 762 of them in its bull state.

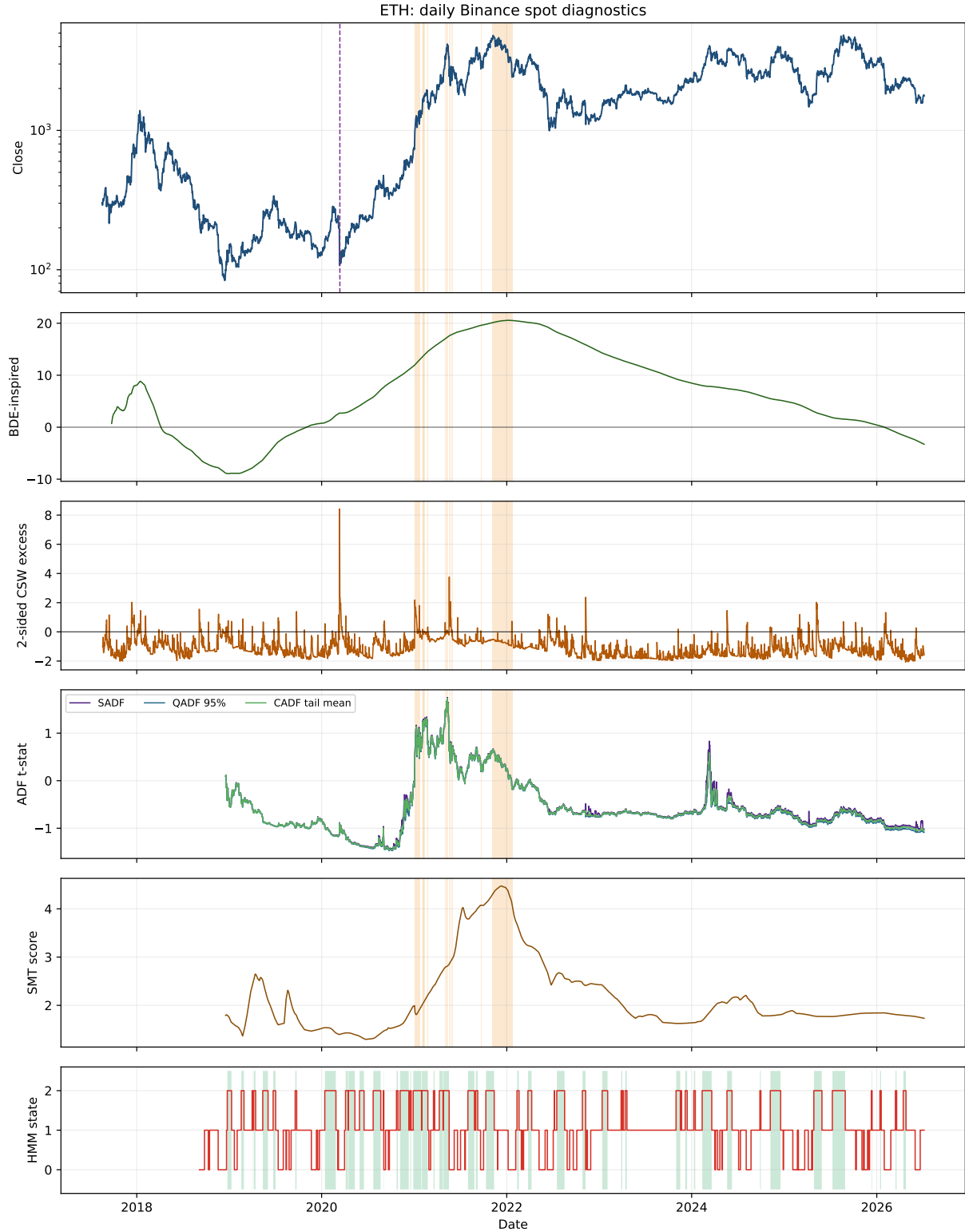


Figure 3: ETH structural-break diagnostics. Orange shading is retrospective consensus, not an online trading signal. Green shading in the bottom panel marks the filtered HMM bull state, which is the online HMM strategy state.

6.3 ETC

6 retrospective consensus alert run(s) were flagged on the daily clock. The first listed run begins on 2021-04-15. The SDFC candidate is none (non-positive t) with t -statistic -0.26. SADF peaks on 2021-05-06, the two-sided CSW heuristic peaks on 2020-03-12, and the SMT score peaks on 2021-11-17. The walk-forward HMM covers 2563 daily observations from 2019-07-02 and spends 505 of them in its bull state.

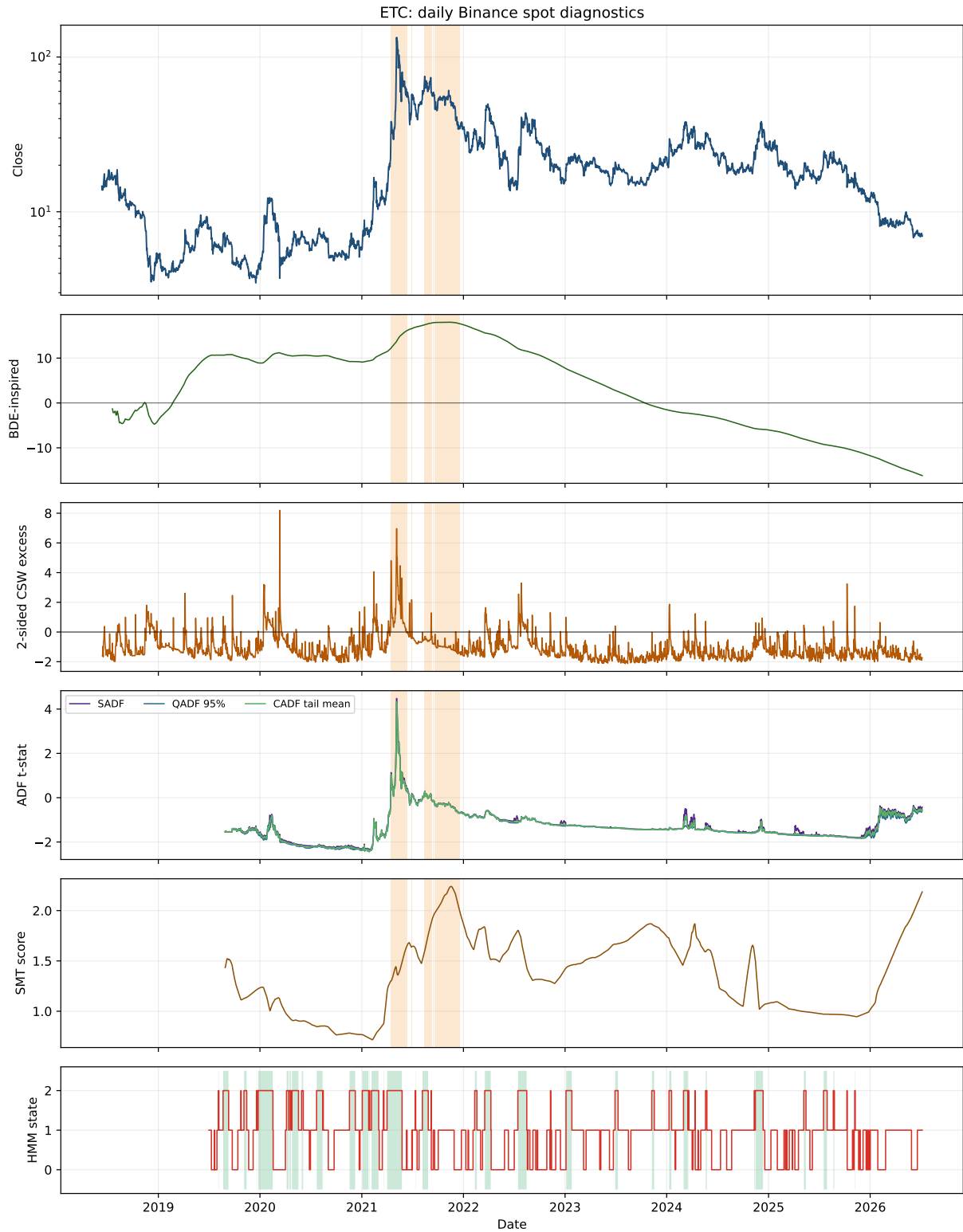


Figure 4: ETC structural-break diagnostics. Orange shading is retrospective consensus, not an online trading signal. Green shading in the bottom panel marks the filtered HMM bull state, which is the online HMM strategy state.

6.4 SOL

10 retrospective consensus alert run(s) were flagged on the daily clock. The first listed run begins on 2021-09-06. The SDFC candidate is 2022-12-30 with t-statistic 0.59. SADF peaks on 2023-12-25,

the two-sided CSW heuristic peaks on 2022-11-09, and the SMT score peaks on 2022-01-04. The walk-forward HMM covers 1772 daily observations from 2021-08-31 and spends 257 of them in its bull state.

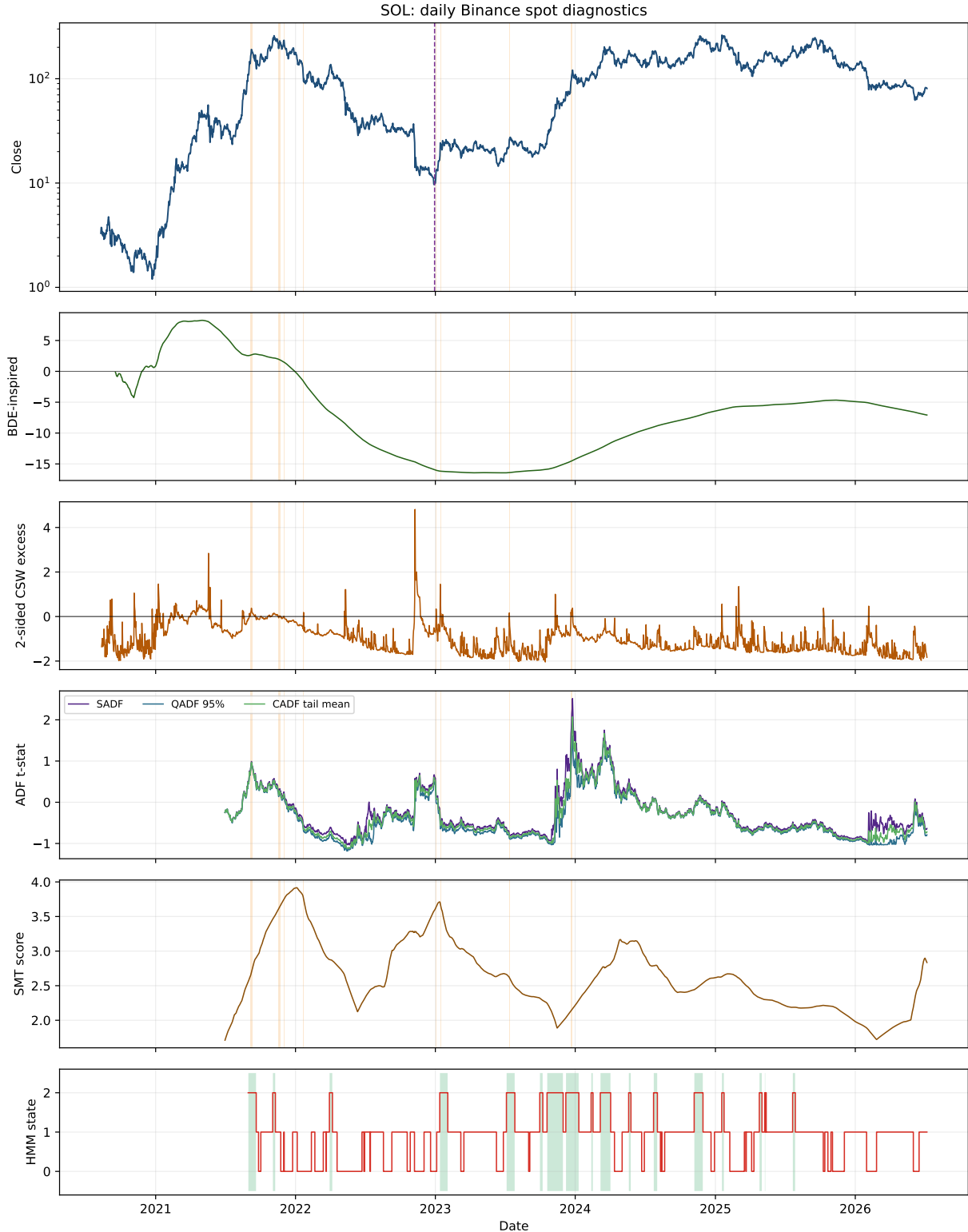


Figure 5: SOL structural-break diagnostics. Orange shading is retrospective consensus, not an online trading signal. Green shading in the bottom panel marks the filtered HMM bull state, which is the online HMM strategy state.

6.5 HYPE

7 retrospective consensus alert run(s) were flagged on the daily clock. The first listed run begins on 2025-12-18. The SDFC candidate is 2026-01-21 with t-statistic 1.71. SADF peaks on 2026-06-01, the two-sided CSW heuristic peaks on 2026-01-28, and the SMT score peaks on 2025-12-28. The walk-forward HMM covers 19 daily observations from 2026-06-19 and spends 1 of them in its bull state. HYPE uses Binance USD-M futures price klines; strategy returns exclude funding, carry, basis, and futures-specific execution costs.

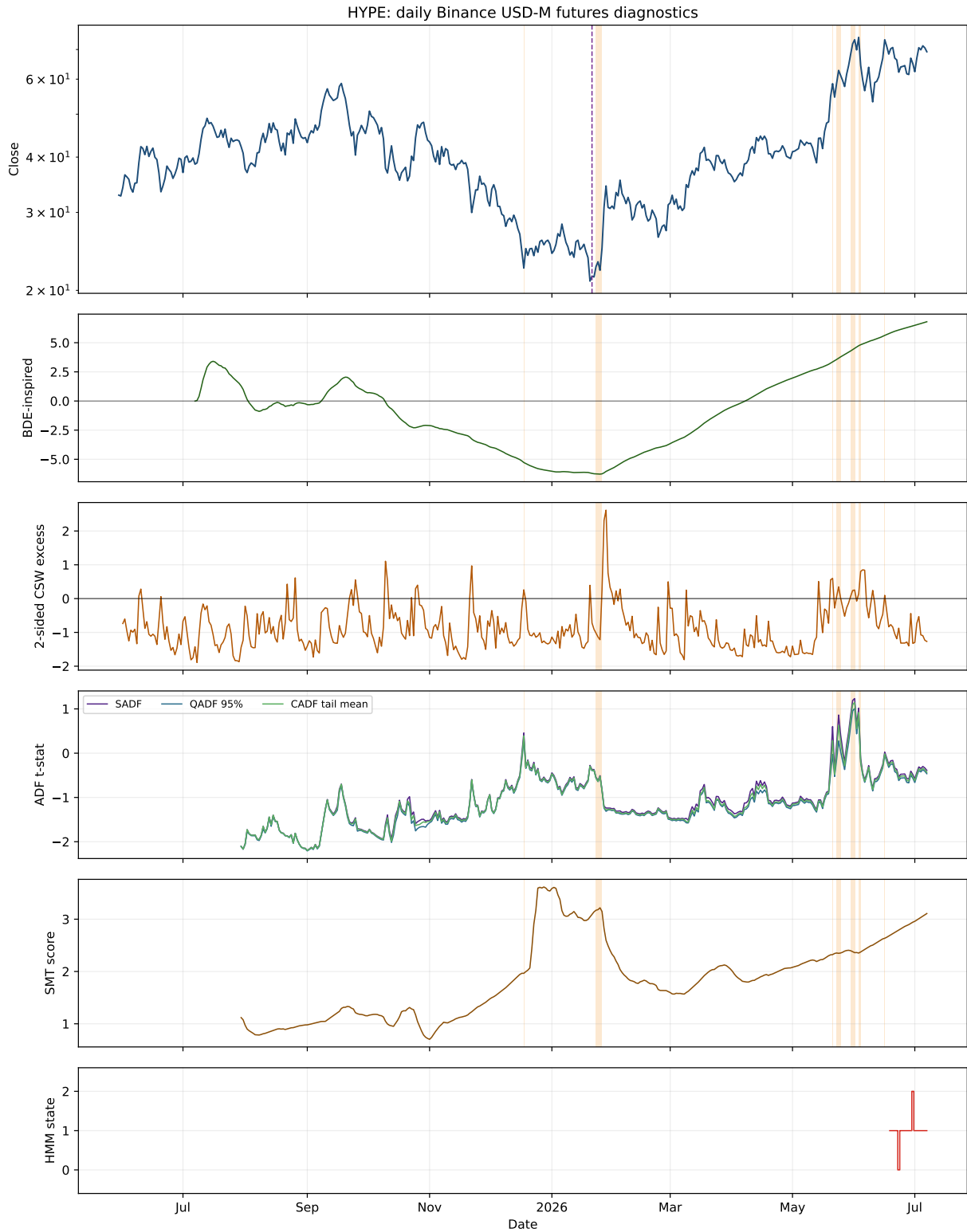


Figure 6: HYPE structural-break diagnostics. Orange shading is retrospective consensus, not an online trading signal. Green shading in the bottom panel marks the filtered HMM bull state, which is the online HMM strategy state.

7 Consensus Alert Periods

The table below lists the first few retrospective periods in which at least two diagnostic families were active. Short alert runs are common because the tests are sensitive to local trend acceleration and volatility normalization.

Table 5: First listed retrospective consensus alert periods. Period length is counted in calendar days between the first and last daily alert observation. For assets with many daily runs, the table shows the first eight runs.

Asset	Start	End	Days
BTC	2021-01-02	2021-01-03	2
	2021-01-05	2021-01-10	6
	2021-01-13	2021-01-14	2
	2021-02-08	2021-02-09	2
	2021-02-11	2021-02-11	1
	2021-02-17	2021-02-23	7
	2021-03-01	2021-03-01	1
	2021-03-11	2021-03-11	1
ETH	2021-01-04	2021-01-11	8
	2021-01-14	2021-01-14	1
	2021-01-16	2021-01-16	1
	2021-01-19	2021-01-20	2
	2021-01-24	2021-01-24	1
	2021-02-03	2021-02-03	1
	2021-02-05	2021-02-05	1
	2021-02-08	2021-02-09	2
ETC	2021-04-15	2021-06-11	58
	2021-06-13	2021-06-14	2
	2021-06-29	2021-06-30	2
	2021-08-12	2021-09-09	29
	2021-09-15	2021-09-15	1
	2021-09-20	2021-12-20	92
SOL	2021-09-06	2021-09-11	6
	2021-11-17	2021-11-17	1
	2021-11-20	2021-11-21	2
	2021-11-23	2021-11-23	1
	2021-12-01	2021-12-02	2
	2022-01-22	2022-01-22	1
	2023-01-03	2023-01-03	1
	2023-01-14	2023-01-16	3
HYPE	2025-12-18	2025-12-18	1
	2026-01-23	2026-01-26	4
	2026-05-21	2026-05-21	1
	2026-05-23	2026-05-25	3
	2026-05-30	2026-06-01	3
	2026-06-03	2026-06-04	2
	2026-06-16	2026-06-16	1

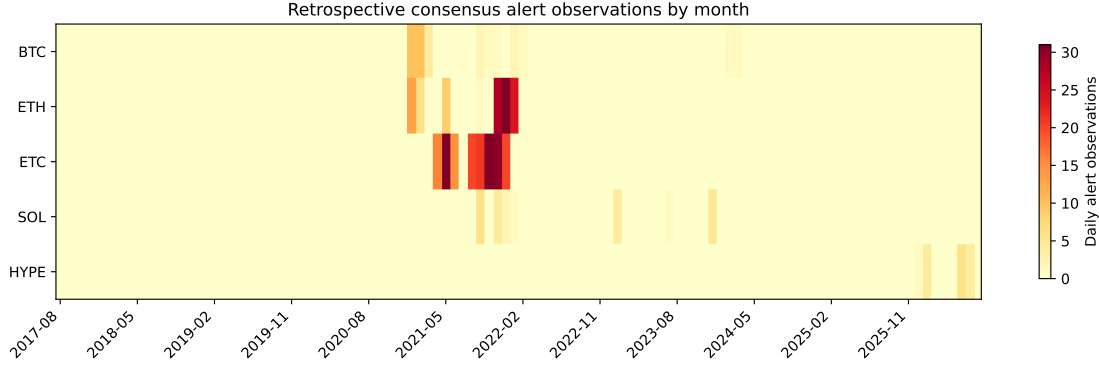


Figure 7: Retrospective consensus alert observations by month.

8 Finite-Sample Calibration and Sensitivity

The table below compares each asset’s observed ADF-family maxima with cached random-walk finite-sample critical values. These simulations give a more defensible reference than the internal top 5% visual flags, although they still inherit the simplified zero-lag ADF specification.

Asset	Sims	τ	SADF 95%	Max SADF	SADF hits	QADF 95%	Max QADF	QADF hits	CADF 95%	Max CADF	CADF hits
BTC	2000	488	2.02	1.89	0	1.92	1.82	0	1.96	1.86	0
ETH	2000	488	2.02	1.75	0	1.92	1.72	0	1.96	1.73	0
ETC	2000	443	2.01	4.48	13	1.90	4.28	14	1.95	4.32	13
SOL	2000	324	2.05	2.52	4	1.94	1.61	0	1.99	2.07	1
HYPE	2000	61	2.03	1.23	0	1.91	1.01	0	1.97	1.14	0

Table 6: Simulated finite-sample critical values under random-walk null paths with matching sample length and τ . Hits count daily observations above the 95% simulated max-statistic threshold.

Table 7: Sensitivity of selected results to the minimum-window fraction. ADF-family windows vary; BDE, CSW, SMT, and HMM inputs are held at their main-run definitions.

Asset	τ rule	τ	Max SADF date	Runs	Best structure signal	Final
BTC	10%	325	2024-03-13	19	CSW	11.68x
BTC	15%	488	2021-01-08	16	CSW	11.68x
BTC	20%	650	2021-02-21	15	CSW	11.68x
ETH	10%	325	2024-03-11	21	BDE	9.35x
ETH	15%	488	2021-05-11	18	BDE	9.35x
ETH	20%	650	2021-05-11	18	BDE	9.35x
ETC	10%	295	2021-05-06	14	HMM	20.48x
ETC	15%	443	2021-05-06	6	HMM	20.48x
ETC	20%	590	2021-05-06	6	HMM	20.48x
SOL	10%	216	2023-12-25	16	CSW	27.43x
SOL	15%	324	2023-12-25	10	CSW	27.43x
SOL	20%	432	2024-03-17	15	CSW	27.43x
HYPE	10%	60	2026-06-01	7	CSW	1.62x
HYPE	15%	61	2026-06-01	7	CSW	1.62x
HYPE	20%	81	2026-06-01	7	CSW	1.62x

9 Long/Flat Backtests

The diagnostics above are research features, not trading rules. Still, it is useful to ask what a simple long-or-cash conversion would have done.

9.1 Backtest Rule

1. At each daily close t , compute diagnostics using data through t only.
2. Convert each eligible indicator into an online alert using prior expanding thresholds, except CSW, whose positive excess is already boundary-based.
3. Require the trailing momentum gate to be positive.
4. An alert computed at close t sets exposure for the close t to close $t + 1$ return, assuming execution at or immediately after close t .
5. Keep the position long for the holding window after a valid alert, then move back out unless a new alert refreshes the holding window. The HMM strategy is the exception: it uses no momentum gate and no holding window, and instead stays long exactly while the filtered bull state is active.
6. Charge transaction costs whenever exposure changes. Buy-and-hold is charged one entry cost for consistency.

This is still an in-sample comparison. The best final PnL is an indicative ranking of which simple signal conversion looked promising on this sample, not evidence of a deployable strategy.

Asset	Best structure signal	Best signal final	Buy-and-hold final	Best including buy-and-hold	Max DD	Exposure	Trades
BTC	CSW	11.68x	14.77x	Buy-and-hold	-37%	17%	20
ETH	BDE	9.35x	5.86x	BDE	-61%	19%	7
ETC	HMM	20.48x	0.48x	HMM	-61%	17%	42
SOL	CSW	27.43x	24.40x	CSW	-60%	20%	9
HYPE	CSW	1.62x	2.11x	Buy-and-hold	-46%	46%	7

Table 8: Best long/flat structure-break strategy by final equity multiple. Best structure signal excludes buy-and-hold; best including buy-and-hold includes it. Max DD, exposure, and trades refer to the best structure signal.

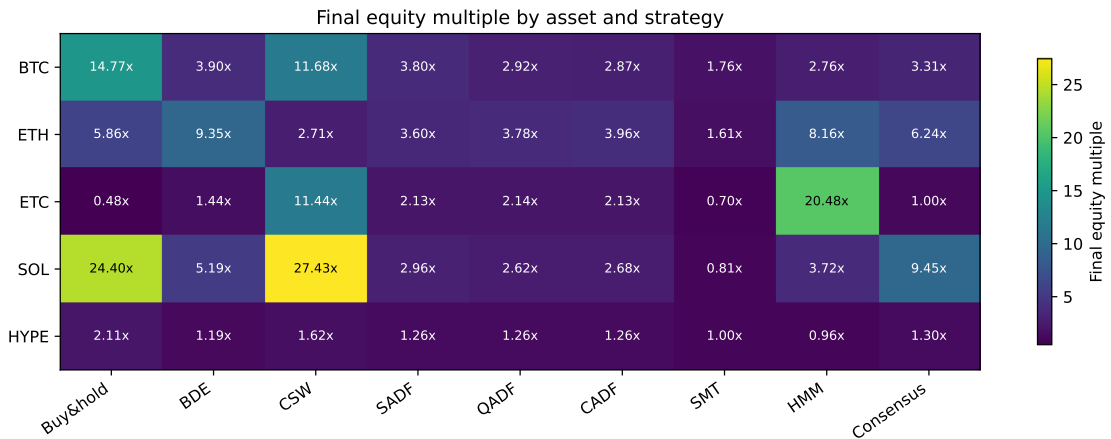


Figure 8: Final equity multiple by asset and strategy. Values are in-sample and normalized to 1 at each asset's first available observation.

9.2 PnL Curves

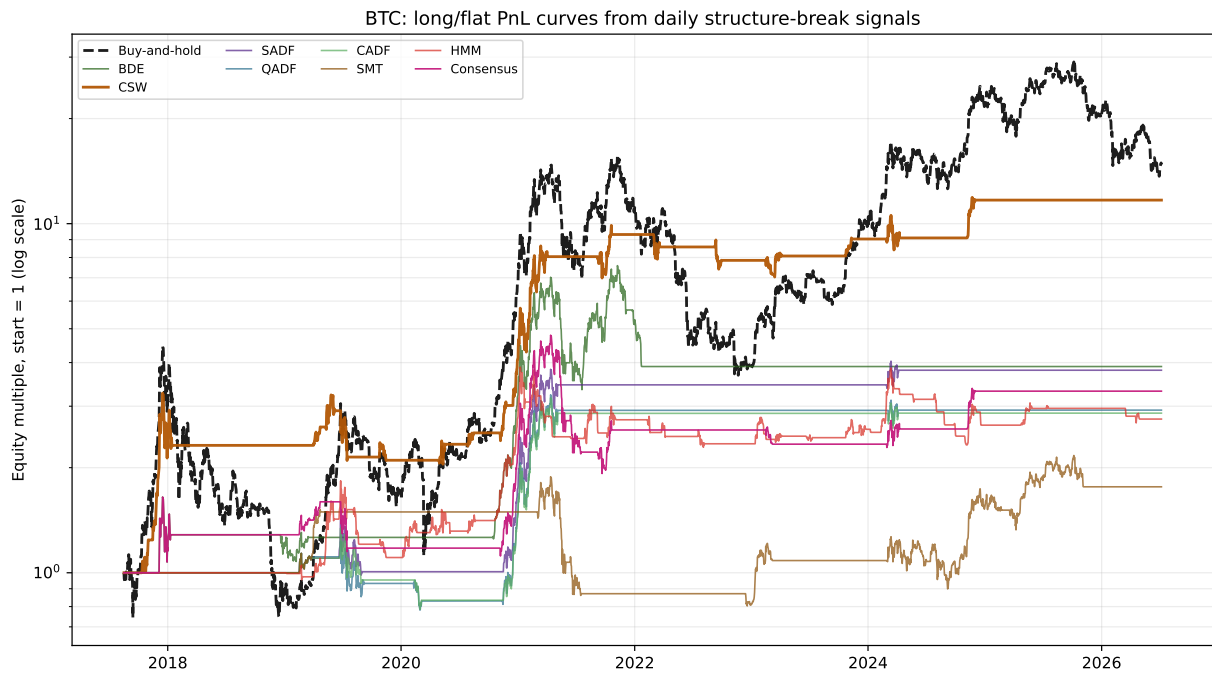


Figure 9: BTC long/flat strategy PnL curves.



Figure 10: ETH long/flat strategy PnL curves.



Figure 11: ETC long/flat strategy PnL curves.

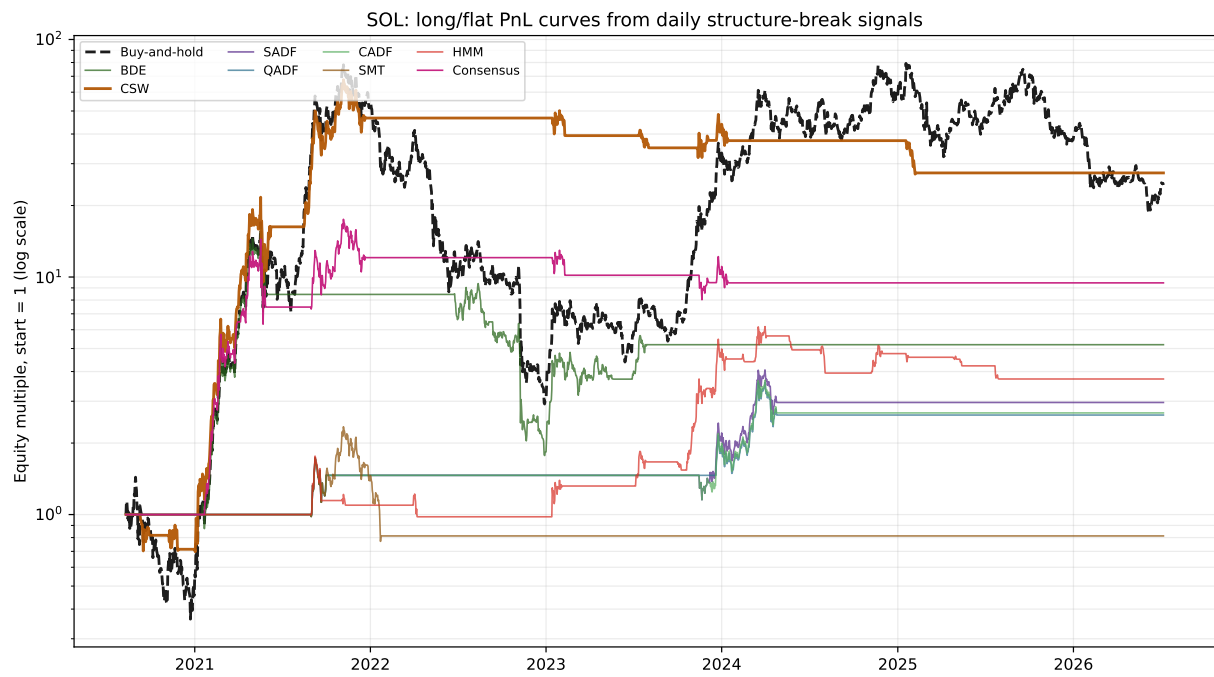


Figure 12: SOL long/flat strategy PnL curves.

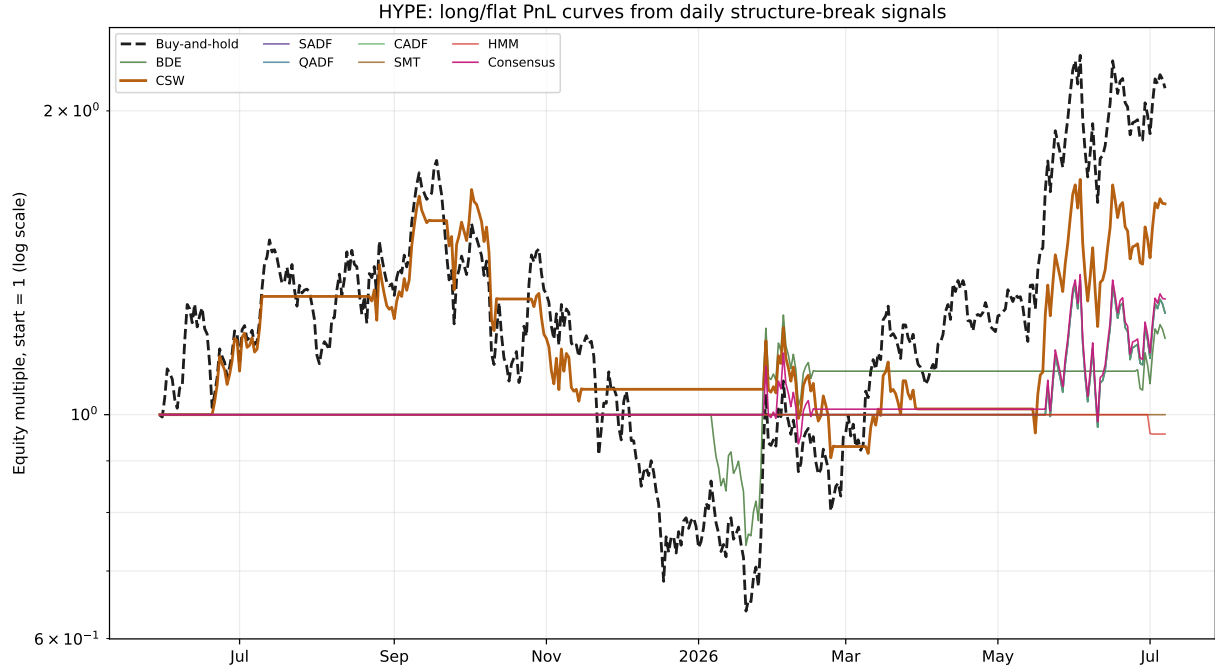


Figure 13: HYPE long/flat strategy PnL curves. HYPE uses Binance USD-M futures price data; returns exclude funding and futures-specific carry.

9.3 Full Strategy Table

Table 9: Simple long/flat strategy performance. Final multiple is the ending equity multiple from 1. CAGR is annualized over each asset’s available sample. Sharpe uses daily returns and 365-day annualization. Because Sharpe is computed from arithmetic daily means, a volatile asset can show a positive Sharpe while its compounded return is negative (volatility drag); ETC buy-and-hold is an example.

Asset	Strategy	Final	Total return	CAGR	Max DD	Sharpe	Exposure	Trades
BTC	Buy-and-hold	14.77x	1377%	35%	-83%	0.79	100%	1
BTC	BDE	3.90x	290%	17%	-52%	0.62	17%	6
BTC	CSW	11.68x	1068%	32%	-37%	0.99	17%	20
BTC	SADF	3.80x	280%	16%	-31%	0.72	10%	5
BTC	QADF	2.92x	192%	13%	-36%	0.60	11%	6
BTC	CADF	2.87x	187%	13%	-41%	0.59	10%	6
BTC	SMT	1.76x	76%	7%	-57%	0.37	26%	11
BTC	HMM	2.76x	176%	12%	-41%	0.53	23%	56
BTC	Consensus	3.31x	231%	14%	-60%	0.57	15%	12
ETH	Buy-and-hold	5.86x	486%	22%	-94%	0.67	100%	1
ETH	BDE	9.35x	835%	29%	-61%	0.76	19%	7
ETH	CSW	2.71x	171%	12%	-64%	0.47	18%	22
ETH	SADF	3.60x	260%	16%	-52%	0.62	6%	2
ETH	QADF	3.78x	278%	16%	-50%	0.65	5%	1
ETH	CADF	3.96x	296%	17%	-50%	0.67	5%	1
ETH	SMT	1.61x	61%	6%	-56%	0.33	11%	5
ETH	HMM	8.16x	716%	27%	-50%	0.75	23%	47
ETH	Consensus	6.24x	524%	23%	-56%	0.68	16%	7
ETC	Buy-and-hold	0.48x	-52%	-9%	-95%	0.40	100%	1
ETC	BDE	1.44x	44%	5%	-74%	0.34	15%	7

Asset	Strategy	Final	Total return	CAGR	Max DD	Sharpe	Exposure	Trades
ETC	CSW	11.44x	1044%	35%	-67%	0.77	25%	24
ETC	SADF	2.13x	113%	10%	-61%	0.42	5%	3
ETC	QADF	2.14x	114%	10%	-61%	0.42	5%	3
ETC	CADF	2.13x	113%	10%	-61%	0.42	5%	3
ETC	SMT	0.70x	-30%	-4%	-84%	0.10	8%	6
ETC	HMM	20.48x	1948%	45%	-61%	0.89	17%	42
ETC	Consensus	1.00x	0%	0%	-78%	0.25	13%	10
SOL	Buy-and-hold	24.40x	2340%	72%	-96%	1.04	100%	1
SOL	BDE	5.19x	419%	32%	-89%	0.75	22%	6
SOL	CSW	27.43x	2643%	75%	-60%	1.14	20%	9
SOL	SADF	2.96x	196%	20%	-35%	0.72	9%	3
SOL	QADF	2.62x	162%	18%	-35%	0.69	7%	2
SOL	CADF	2.68x	168%	18%	-35%	0.67	9%	3
SOL	SMT	0.81x	-19%	-3%	-67%	0.03	7%	2
SOL	HMM	3.72x	272%	25%	-44%	0.74	12%	18
SOL	Consensus	9.45x	845%	46%	-56%	0.94	14%	6
HYPE	Buy-and-hold	2.11x	111%	96%	-64%	1.17	100%	1
HYPE	BDE	1.19x	19%	17%	-26%	0.58	13%	2
HYPE	CSW	1.62x	62%	55%	-46%	0.99	46%	7
HYPE	SADF	1.26x	26%	23%	-29%	0.75	12%	1
HYPE	QADF	1.26x	26%	23%	-29%	0.75	12%	1
HYPE	CADF	1.26x	26%	23%	-29%	0.75	12%	1
HYPE	SMT	1.00x	0%	0%	0%	n/a	0%	0
HYPE	HMM	0.96x	-4%	-4%	-4%	-0.97	0%	1
HYPE	Consensus	1.30x	30%	27%	-29%	0.76	17%	2

9.4 Walk-Forward and Multiple-Testing Checks

The walk-forward check selects the best structure signal on an expanding training window and evaluates only the next out-of-sample block. This does not prove profitability, but it reduces the most obvious winner-picking bias in the in-sample table. In the selected-signal path, repeated selections are compressed, so “CSW x6” means CSW was chosen for six consecutive blocks.

Asset	Blocks	OOS days	Selected signal path	OOS final	OOS Sharpe	OOS Max DD
BTC	13	2340	CSW x13	5.45x	1.08	-29%
ETH	13	2340	BDE x13	5.36x	0.79	-61%
ETC	12	2160	CSW x6, HMM, CSW x2, HMM x3	6.45x	0.78	-61%
SOL	7	1260	CSW x7	0.59x	-0.35	-46%
HYPE	0	0	n/a	n/a	n/a	n/a

Table 10: Expanding-window walk-forward validation of signal selection. HYPE has too little history for the default train/test split and may show no blocks.

Asset	Bootstrap sims	Best signal	Best final	Final 5–95% CI	Sharpe 5–95% CI	Reality-check p
BTC	500	CSW	11.68x	[2.53x, 83.04x]	[0.49, 1.56]	0.946
ETH	500	BDE	9.35x	[0.96x, 81.09x]	[0.21, 1.23]	0.908
ETC	500	HMM	20.48x	[1.11x, 480.46x]	[0.27, 1.44]	0.512
SOL	500	CSW	27.43x	[1.77x, 637.25x]	[0.45, 1.81]	0.958
HYPE	500	CSW	1.62x	[0.61x, 4.16x]	[-0.38, 2.31]	0.910

Table 11: Block-bootstrap uncertainty for the best in-sample structure signal and a simple centered bootstrap reality-check p -value for best excess mean return versus buy-and-hold.

The reality-check p -value estimates how often a resampled, mean-centered version of the signal family would produce a best mean daily excess return over buy-and-hold at least as large as the one observed, after accounting for the fact that the best of several signals was selected. Values near 1 mean the observed outperformance is entirely consistent with luck; only small values (for example below 0.05) would count as evidence of skill. Readers should check the table rather than assume the headline equity multiples imply significance.

10 Interpretation

Methodologically, SADF is a strong default diagnostic for repeated explosive episodes because it does not require the analyst to know the number or timing of regime switches. Empirically, in this particular in-sample long/flat conversion, the winning signal can differ by asset and must be treated as selected after looking at many alternatives.

The most important traps are multiple testing, listing-history differences, simplified zero-lag ADF regressions, and price-only data. HYPE is especially limited because it uses futures price klines without funding-rate adjustment. The HMM adds a complementary latent-state view, but its in-sample equity multiples share the same selection caveats, and its output is sensitive to the number of states and the refit schedule. Regime-change diagnostics are best read as feature-engineering tools, not as a recipe for direct trading.

11 Minimal Reproducibility Sketch

```
# online execution convention used in the backtest
event_t = signal_at_close_t
position = event_t.rolling(hold_days).max().shift(1)
strategy_return_t = position_t * close.pct_change_t - costs_t

# HMM variant: no hold window, position tracks the filtered bull state
position_hmm = (filtered_state_at_close_t == bull_state).shift(1)
```

12 Reproduce

```
python3 scripts/generate_daily_regime_chapter.py --use-cache
pdflatex -interaction=nonstopmode regime_detection_crypto_report.tex
pdflatex -interaction=nonstopmode regime_detection_crypto_report.tex
```

The generator writes daily diagnostics, strategy PnL CSVs, cached calibration files, strict JSON metadata, all figures, and this TeX report. The rendered PDF is then produced by `pdflatex`.

13 References

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- Binance API daily kline endpoints for BTCUSDT, ETHUSDT, ETCUSDT, SOLUSDT spot, and HYPEUSDT USD-M futures fallback.
- Local reproduction script: `scripts/generate_daily_regime_chapter.py`.